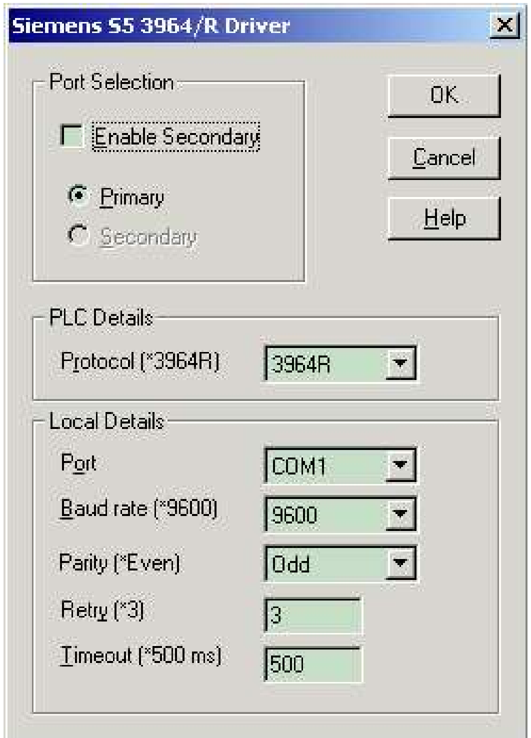
Chapter 14 : Siemens SIE3964R protocol driver

Overview

Adroit supports the Siemens S5 3964/R driver protocol.



PLC Details

Protocol - 3964 No BCC checking, 3964R BCC checking.

Local Details

Port - The available serial ports on your machine

Baud Rate - The speed of data transmission. Any of the following Baud Rates may be specified: 2400, 4800, 9600, or 19200.

Parity – (Default even) Select none, odd or even as required.

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Wiring

PLC 25 pin		PC 25 pin
1	Shield	1
2		3
3		2
7		7

Note: *Connect shield to PLC side only*

PLC software

PLC software is required to allow communication with the CP524 and CP525 cards.

Supported addresses

Siemens S5 PLC data address types supported by the SIE3964R Adroit Protocol driver Version 1.0

Device		Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STR
Output Bytes	1	OB 0.0	OB 255.7	Digital	X			
Input Bytes	2	IB 0.0	IB 255.7	Digital	X			
Flags Bytes	3	FB 0.0	FB 255.7	Digital	X			
Timers	4	ZB 0	ZB 255	Signed Integer	N	Y	Y	N
Counters	5	CB 0	ZB 255	Signed Integer	N	Y	Y	N
Extended DB	6	DX 0:DW0.0	DX255:DW 255.15	Digital	Y	N	N	N
Extended DB	7	DX 0:DW0	DX255:DW 255	Signed Integer	N	Y	Y	N
Data Block	8	DB 0:DW 0.0	DB 255:DW 255.15	Digital	X			
	9	DB 0:DW 0	DB 255:DW 255	Signed Integer		X	X	
	10	DB 0:DD 0 KF	DB 255:DD 255 KF	32 Bit Integer		X	X	
	11	DB 0:DD 0 KG	DB 255:DD 255 KG	32 Bit IEEE Float			X	

Message protocol

The SIE3964R protocol involves a number of separate instructions for each memory read or memory write transaction. This is a point-to-point protocol and therefore device addressing is not supported.

The basic sequence of events for any transaction is displayed below. The arrows indicate P (for Protocol Driver) and S (for Siemens) to denote direction of communication. The transaction data may be read or write requests.

The master (protocol driver) commences the transaction with:

Mnemonic	Value	Direction	Explanation
STX	02h	P→S	Hi, are you ready?
DLE	10h	P←S	Yes
[Transaction]	data	P→S	Here's the message
DLE	10h	P→S	Terminator
ETX	03h	P→S	End of message
DLE	10h	P←S	Thanks

Within 5 seconds, the slave (Siemens device) responds with:

Mnemonic	Value	Direction	Explanation
STX	02h	P←S	Hi, are you ready?
DLE	10h	P→S	Yes
[Response]		P←S	Here's my response
DLE	10h	P←S	Terminator
ETX	03h	P←S	End of message
DLE	10h	P→S	Thanks

Follow-on telegrams are not supported/used by the protocol driver.

The rest of this section will concern itself only with the [Transaction] and with the [Response] structures.

The 3964R protocol uses the last byte of these structures as a byte checksum. The 3964 protocol does not use checksums. The checksum capability is configured per SIE3964R connection device in the protocol driver setup procedure.

The 3964/3964R protocol employs DLE doubling in these structures whereby any DLE that forms part of the data is padded immediately with another DLE.

Output flag bytes 0-255

Read request:

		Fetch	Data Type	Source H	Source L	Number H	Number L	CF	CPU
00	00	45	41	xx	xx	xx	xx	00	00

Read response:

			Error num	1st data byte	nth data byte
0	0	0	0	xx	xx

Write request:

Not allowed

Input flag bytes 0-255

Read request:

		Fetch	Data Type	Source H	Source L	Number H	Number L	CF	CPU
00	00	45	45	xx	xx	xx	xx	00	00

Read response:

			Error num	1st data byte	nth data byte
0	0	0	0	xx	xx

Write request:

Not allowed

Flag bytes 0-255

Read request:

		Fetch	Data Type	Source H	Source L	Number H	Number L	CF	CPU
00	00	45	4D	xx	xx	xx	xx	00	00

Read response:

			Error num	1st data byte	nth data byte
0	0	0	0	xx	xx

Write request:

Not allowed

Data blocks 0-255, data words 0-255

Read request:

		Fetch	Data Type	DB Num	DW Num	Number H	Number L	CF	CPU
00	00	45	44	xx	xx	xx	xx	00	00

Read response:

			Error num	1st data byte	nth data byte
0	0	0	0	xx	xx

Write request:

		Write	Data Type	DB Num	DW Num	Number H	Number L	CF	CPU
00	00	41	44	xx	xx	xx	xx	00	00

1st data byte	nth data byte
xx	xx

Write response:

			Error num
0	0	0	0

Document Revisions

Date	Changes
25-02-2003	New screenshot, showing parity selection option, fixed wirrring diagram characters to be lines.